

Srajan

✉ srajan.ai24@bmsce.ac.in 📞 +91 9482351696

🐙 github.com/srajan-kumar1 | 🔗 linkedin.com/in/srajan-kumar

Profile

Artificial Intelligence and Machine Learning undergraduate with strong foundations in machine learning, inferential statistics, computer vision, and full-stack development. Experienced in building intelligent software systems, analyzing real-world datasets, and developing data-driven solutions using Python and modern web technologies. Passionate about solving practical engineering problems through AI while continuously improving analytical, research, and software development skills.

Technical Skills

Programming Languages: C, Java, Python

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-Learn, OpenCV, Seaborn, SciPy, FastAPI

Web Technologies: HTML, CSS, JavaScript

Databases: SQL, MongoDB

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, IntelliJ IDEA, Optuna

Education

B.M.S. College of Engineering, Bengaluru
B.E. in Artificial Intelligence and Machine Learning
12th – JIPUC Kundapura
10th – St. Thomas HPS Byndoor

2024 – 2028
CGPA: **9.19/10**
95.33%
93.20%

Projects

Market Prediction System Using Machine Learning *Python, FastAPI, Scikit-Learn, Pandas, NumPy*

- Developed an AI-powered market prediction system capable of generating real-time BUY, SELL, and HOLD signals for stocks, forex, cryptocurrencies, and commodities.
- Engineered technical indicators including SMA, EMA, RSI, volatility, and lag-based features to improve prediction accuracy from historical market data.
- Implemented multiple machine learning models including KNN, Decision Tree, Random Forest, Bagging, and AdaBoost with Optuna-based hyperparameter optimization.
- Built a FastAPI backend with WebSocket support for real-time prediction streaming, dashboard visualization, and strategy backtesting.

Employee Attrition Prediction using Machine Learning and Inferential Statistics *Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy*

- Developed a machine learning system to predict employee attrition using the IBM HR Analytics Employee Attrition dataset consisting of 1,470 employee records and 35 features.
- Compared four machine learning models—Logistic Regression, Random Forest, XGBoost, and Support Vector Machine (SVM)—using Stratified K-Fold, Bootstrap Sampling (with replacement), and Random K-Fold (without replacement).
- Evaluated the impact of different sampling strategies on model performance using inferential statistical techniques including Paired t-Test, Chi-Square Test, and Shapiro-Wilk Test.
- Addressed class imbalance using SMOTE and analyzed its influence on model accuracy, statistical significance, and reliability of sampling techniques.
- Visualized experimental results through grouped accuracy charts, box plots, feature significance graphs, and comparative model evaluation plots to support statistical conclusions.

Relevant Coursework

Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Computer Organization, Probability and Statistics, Inferential Statistics, Machine Learning Fundamentals, Artificial Intelligence.

Achievements

- Solved over **150+** Data Structures and Algorithms problems on LeetCode.
- Maintained an academic CGPA of **9.19/10** in the Artificial Intelligence and Machine Learning program.